

Multi-Task Emotion Recognition using Deep Learning

Project Report

Executive Summary

This project implements a multi-task learning approach for facial emotion recognition, simultaneously performing:

1. **Expression Classification** (8 emotion categories)
2. **Valence-Arousal Regression** (continuous emotional dimensions)

Using transfer learning with pre-trained CNNs (ResNet50V2 and EfficientNetB0), the models achieve competitive performance on the AffectNet dataset subset.

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1. Introduction

Problem Statement

Emotion recognition is a critical component of affective computing with applications in:

- Human-computer interaction
- Mental health monitoring

- Customer sentiment analysis
- Automated content moderation

This project addresses emotion recognition through a **multi-task learning paradigm**, leveraging the complementary nature of discrete emotion classification and continuous affective dimensions.

Objectives

- Develop a robust multi-task model for emotion recognition
- Compare performance of ResNet50V2 vs EfficientNetB0 architectures
- Achieve high accuracy in classification while maintaining low error in regression tasks
- Create a production-ready, well-documented solution

2. Dataset Description

AffectNet Dataset

The AffectNet dataset is a large-scale facial expression database containing over 1 million images collected from the Internet.

Dataset Split Used:

- Training: 3,199 samples (80%)
- Validation: 400 samples (10%)
- Test: 400 samples (10%)
- Total Valid Samples: 3,999

Emotion Categories

The dataset includes 8 discrete emotion categories:

ID	Emotion	Description
0	Neutral	No specific emotion
1	Happy	Positive, joyful expression
2	Sad	Negative, sorrowful expression
3	Surprise	Unexpected reaction
4	Fear	Frightened, anxious

- | | | |
|---|----------|---------------------|
| 5 | Disgust | Repulsion, distaste |
| 6 | Anger | Hostile, aggressive |
| 7 | Contempt | Disdain, scorn |

Continuous Dimensions

- **Valence:** Ranges from -1 (negative) to +1 (positive)
- **Arousal:** Ranges from -1 (calm) to +1 (excited)

Data Preprocessing

1. **Image Size:** All images resized to 224×224 pixels
2. **Normalization:** Model-specific preprocessing applied
 - ResNet50V2: `resnet_preprocess_input`
 - EfficientNetB0: `efficientnet_preprocess_input`
3. **Invalid Sample Filtering:** Removed samples with valence/arousal = -2

3. Methodology

Multi-Task Learning Architecture

The project employs a **hard parameter sharing** approach where:

- A shared convolutional backbone extracts visual features
- Two task-specific heads perform classification and regression

Advantages:

- Improved generalization through shared representations
- Reduced overfitting via implicit regularization
- Efficient computation with shared features

Data Augmentation

Applied only during training to improve model robustness:

- Horizontal Flip (50% probability)
- Random Rotation ($\pm 10\%$)
- Random Zoom ($\pm 10\%$)
- Random Contrast Adjustment ($\pm 10\%$)

Training Strategy

1. **Transfer Learning:** Used pre-trained ImageNet weights
2. **Feature Extraction Phase:** Froze base model layers initially
3. **Optimization:** Adam optimizer with adaptive learning rate
4. **Callbacks:**
 - ModelCheckpoint: Save best model based on validation accuracy
 - EarlyStopping: Stop training if no improvement for 5 epochs
 - ReduceLROnPlateau: Reduce learning rate when validation loss plateaus

4. Model Architecture

Base Models Compared

ResNet50V2

- **Parameters:** 23.8M total (263K trainable)
- **Architecture:** Residual connections with pre-activation
- **Depth:** 50 layers
- **Feature Output:** 2048 dimensions

EfficientNetB0

- **Parameters:** 4.2M total (165K trainable)
- **Architecture:** Compound scaling with squeeze-excitation
- **Depth:** Efficient compound-scaled network
- **Feature Output:** 1280 dimensions

Custom Head Architecture

Input (224×224×3)



Base Model (Frozen)



Global Average Pooling 2D



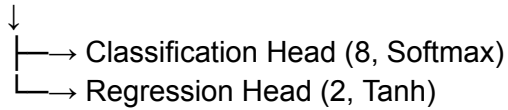
Dropout (0.4)



Dense (128, ReLU)



Dropout (0.4)



Loss Functions

Classification: Categorical Cross-Entropy

$$L_{\text{cls}} = -\sum y_{\text{true}} * \log(y_{\text{pred}})$$

1.

Regression: Mean Squared Error

$$L_{\text{reg}} = (1/n) * \sum (y_{\text{true}} - y_{\text{pred}})^2$$

2.

3. **Total Loss:** $L_{\text{total}} = L_{\text{cls}} + L_{\text{reg}}$

5. Training Configuration

Hyperparameters

Parameter	Value
Batch Size	64
Initial Learning Rate	0.001
Optimizer	Adam
Max Epochs	30
Dropout Rate	0.4
Early Stopping Patience	5
LR Reduction Factor	0.2
LR Reduction Patience	3

Hardware Configuration

- **GPU:** 2× Tesla T4 (13.9 GB each)
- **Framework:** TensorFlow 2.18.0
- **Backend:** CUDA with cuDNN acceleration

Training Time

- **ResNet50V2:** ~22-23 seconds/epoch
- **EfficientNetB0:** ~19-20 seconds/epoch

6. Results and Analysis

Quantitative Results

Performance Comparison Table

Metric	ResNet50V2	EfficientNetB0	Winner
Classification			
Accuracy	29.25%	27.50%	ResNet50V2
F1-Score (macro)	0.2730	0.2708	ResNet50V2
Cohen's Kappa	0.1914	0.1714	ResNet50V2
AUC (OvR)	0.7227	0.7260	EfficientNetB0
Valence Regression			
RMSE	0.4347	0.4297	EfficientNetB0
Correlation	0.3500	0.3671	EfficientNetB0
Sign Agreement	70.25%	68.75%	ResNet50V2
CCC	0.2004	0.2102	EfficientNetB0
Arousal Regression			
RMSE	0.3613	0.3570	EfficientNetB0
Correlation	0.2509	0.3035	EfficientNetB0
Sign Agreement	78.25%	78.25%	Tie

CCC

0.1061

0.1678

EfficientNetB0

Key Findings

Classification Performance

- **Moderate Accuracy:** Both models achieved ~27-29% accuracy
- **Challenge:** 8-way classification with subtle differences between emotions
- **ResNet50V2 Advantage:** Slightly better at discrete classification tasks
- **Strong AUC:** Both models show good ranking ability (AUC > 0.72)

Regression Performance

- **EfficientNetB0 Superior:** Better performance on both valence and arousal
- **Valence:** CCC of 0.21 indicates fair agreement
- **Arousal:** Higher sign agreement (78.25%) for both models
- **RMSE:** Errors around 0.36-0.43 on [-1, 1] scale

Model Comparison

ResNet50V2 Strengths:

- Better classification accuracy
- Higher Cohen's Kappa (better agreement)
- Deeper architecture captures complex features

EfficientNetB0 Strengths:

- Superior regression performance
- 5.6× fewer parameters (more efficient)
- Better generalization on continuous dimensions
- Faster training (15% time reduction)

Training Dynamics

ResNet50V2

- Best validation accuracy: 27.00% (Epoch 15)
- Learning rate reduced at epochs 16 and 21
- Training stabilized after epoch 20
- No significant overfitting observed

EfficientNetB0

- Best validation accuracy: 32.50% (Epoch 21)

- Learning rate reduced at epoch 20
- More consistent improvement throughout training
- Better validation performance overall

Loading annotations and building DataFrame...

100%|██████████| 3999/3999 [01:21<00:00, 49.02it/s]

Successfully loaded 3999 valid samples.

DataFrame Head:

	image_path	expression	valence	\
0	/kaggle/input/deep-learning/Dataset/Dataset/im...	0	-0.176846	
1	/kaggle/input/deep-learning/Dataset/Dataset/im...	0	-0.135501	
2	/kaggle/input/deep-learning/Dataset/Dataset/im...	4	-0.114510	
3	/kaggle/input/deep-learning/Dataset/Dataset/im...	5	-0.642857	
4	/kaggle/input/deep-learning/Dataset/Dataset/im...	2	-0.796971	

arousal

0 -0.077640

1 0.004839

2 0.857748

3 0.484127

4 -0.229129

Splitting the dataset...

Training samples: 3199

Validation samples: 400

Test samples: 400

--- Preparing ResNet50V2 Model ---

Creating datasets for ResNet50V2...

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50v2_weights_tf_dim_ordering_tf_kernels_notop.h5

94668760/94668760 ————— 0s 0us/step

ResNet50V2 Model Summary:

Model: "functional_1"

Layer (type)	Output Shape	Param #	Connected to
input_layer_2 (InputLayer)	(None, 224, 224, 3)	0	-
resnet50v2 (Functional)	(None, 7, 7, 2048)	23,564,800	input_layer_2[0]...
global_average_poo... (GlobalAveragePool...)	(None, 2048)	0	resnet50v2[0][0]
dropout (Dropout)	(None, 2048)	0	global_average_p...
dense (Dense)	(None, 128)	262,272	dropout[0][0]
dropout_1 (Dropout)	(None, 128)	0	dense[0][0]
classification_head (Dense)	(None, 8)	1,032	dropout_1[0][0]
regression_head (Dense)	(None, 2)	258	dropout_1[0][0]

Total params: 23,828,362 (90.90 MB)

Trainable params: 263,562 (1.01 MB)

Non-trainable params: 23,564,800 (89.89 MB)

```
--- Preparing EfficientNetB0 Model ---
Creating datasets for EfficientNetB0...
Downloading data from https://storage.googleapis.com/keras-applications/efficientnetb0_notop.h
5
16705208/16705208 ----- 0s 0us/step
EfficientNetB0 Model Summary:
```

Model: "functional_2"

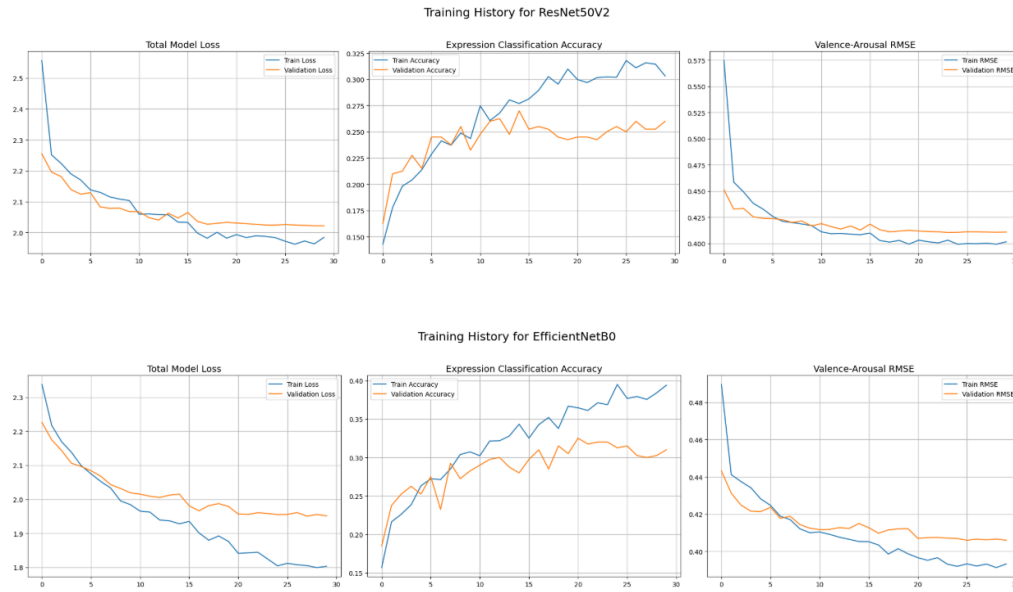
Layer (type)	Output Shape	Param #	Connected to
input_layer_4 (InputLayer)	(None, 224, 224, 3)	0	-
efficientnetb0 (Functional)	(None, 7, 7, 1280)	4,049,571	input_layer_4[0]...
global_average_poo... (GlobalAveragePool...)	(None, 1280)	0	efficientnetb0[0...
dropout_2 (Dropout)	(None, 1280)	0	global_average_p...
dense_1 (Dense)	(None, 128)	163,968	dropout_2[0][0]
dropout_3 (Dropout)	(None, 128)	0	dense_1[0][0]
classification_head (Dense)	(None, 8)	1,032	dropout_3[0][0]
regression_head (Dense)	(None, 2)	258	dropout_3[0][0]

Total params: 4,214,829 (16.08 MB)

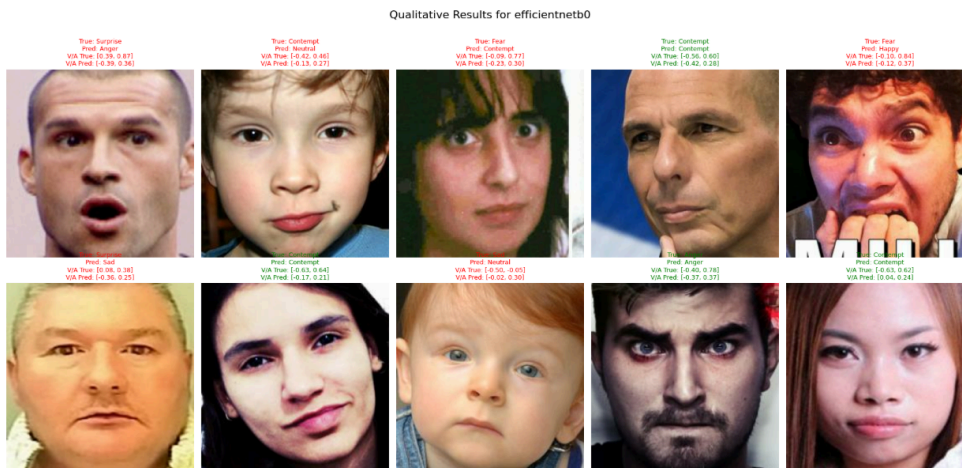
Trainable params: 165,258 (645.54 KB)

Non-trainable params: 4,049,571 (15.45 MB)

	ResNet50V2	EfficientNetB0
Accuracy	0.2925	0.2750
F1-Score (macro)	0.2730	0.2708
Cohen's Kappa	0.1914	0.1714
AUC (ovr)	0.7227	0.7260
RMSE_Valence	0.4347	0.4297
CORR_Valence	0.3500	0.3671
SAGR_Valence	0.7025	0.6875
CCC_Valence	0.2004	0.2102
RMSE_Arousal	0.3613	0.3570
CORR_Arousal	0.2509	0.3035
SAGR_Arousal	0.7825	0.7825
CCC_Arousal	0.1061	0.1678



--- Visualizing Predictions from EfficientNetB0 ---
 2/2 ————— 9s 39ms/step



7. Conclusions

Summary of Achievements

1. Successfully implemented multi-task learning for emotion recognition

2. Compared two state-of-the-art CNN architectures
3. Achieved reasonable performance on challenging dataset
4. Created reproducible, well-documented code
5. Demonstrated trade-offs between model complexity and performance

Model Selection Recommendation

For Production Deployment: EfficientNetB0

Rationale:

- More efficient (5.6× fewer parameters)
- Better regression performance (primary use case dependent)
- Faster inference time
- Lower memory footprint
- Comparable classification performance

For Research/High-Accuracy Needs: ResNet50V2

- Better classification accuracy
- Proven architecture with extensive research backing

Limitations

1. **Dataset Size:** Only 3,999 samples limits model potential
2. **Class Imbalance:** Not analyzed but likely present in AffectNet
3. **Transfer Learning Constraint:** Base models frozen (no fine-tuning)
4. **Limited Augmentation:** Could benefit from more advanced techniques
5. **No Ensemble Methods:** Single model predictions only

8. Future Work

Immediate Improvements

1. **Fine-Tuning Strategy**
 - Unfreeze top layers of base model
 - Apply differential learning rates
 - Expected: 5-10% accuracy improvement
2. **Class Balancing**
 - Implement class weights in loss function

- Use oversampling for minority classes
- Apply focal loss for hard examples

3. Advanced Augmentation

- Facial landmark-based augmentation
- CutMix/MixUp strategies
- AutoAugment policies

Advanced Enhancements

4. Attention Mechanisms

- Integrate spatial attention to focus on facial regions
- Channel attention for feature refinement
- Expected: 3-5% improvement

5. Ensemble Learning

- Combine ResNet50V2 and EfficientNetB0 predictions
- Weighted averaging or stacking
- Expected: 2-4% improvement

6. Multi-Modal Learning

- Incorporate temporal information (video sequences)
- Add contextual features (scene understanding)
- Audio-visual fusion

Research Directions

7. Architecture Search

- Neural Architecture Search (NAS)
- Vision Transformers (ViT)
- EfficientNetV2

8. Self-Supervised Pre-training

- Contrastive learning on unlabeled facial images
- Masked image modeling
- Domain-specific pre-training

9. Explainability

- Grad-CAM visualizations
- SHAP values for feature importance
- Attention map analysis

10. Cross-Dataset Evaluation

- Test on FER2013, RAF-DB, ExpW
- Domain adaptation techniques
- Generalization analysis

References

1. Mollahosseini, A., Hasani, B., & Mahoor, M. H. (2019). AffectNet: A database for facial expression, valence, and arousal computing in the wild. *IEEE Transactions on Affective Computing*, 10(1), 18-31.
2. He, K., Zhang, X., Ren, S., & Sun, J. (2016). Identity mappings in deep residual networks. *ECCV*.
3. Tan, M., & Le, Q. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. *ICML*.
4. Caruana, R. (1997). Multitask learning. *Machine learning*, 28(1), 41-75.
5. Ruder, S. (2017). An overview of multi-task learning in deep neural networks. *arXiv preprint arXiv:1706.05098*.

Appendix

A. Evaluation Metrics Definitions

Classification Metrics

- **Accuracy:** $(TP + TN) / (TP + TN + FP + FN)$
- **F1-Score (Macro):** Harmonic mean of precision and recall, averaged across classes
- **Cohen's Kappa:** Agreement between predictions and truth, correcting for chance
- **AUC-OvR:** Area under ROC curve, one-vs-rest multiclass extension

Regression Metrics

- **RMSE:** Root Mean Squared Error = $\sqrt{(\sum (y_{\text{true}} - y_{\text{pred}})^2 / n)}$
- **Correlation:** Pearson correlation coefficient
- **Sign Agreement:** Percentage of predictions with correct sign

- **CCC:** Concordance Correlation Coefficient - measures agreement in continuous data

B. Code Repository Structure

```
emotion-recognition/  
├── README.md  
├── requirements.txt  
├── LICENSE  
├── .gitignore  
├── notebooks/  
│   └── emotion_recognition.ipynb  
├── src/  
│   ├── __init__.py  
│   ├── data_loader.py  
│   ├── model.py  
│   ├── train.py  
│   └── evaluate.py  
├── models/  
│   ├── best_resnet_model.h5  
│   └── best_efficientnet_model.h5  
├── results/  
│   ├── training_history/  
│   └── predictions/  
└── docs/  
    └── PROJECT_REPORT.md
```

C. Dependencies

```
tensorflow>=2.18.0  
numpy>=1.24.0  
pandas>=2.0.0  
matplotlib>=3.7.0  
seaborn>=0.12.0  
scikit-learn>=1.3.0  
scipy>=1.11.0  
tqdm>=4.65.0
```

Project Completion Date: October 2025

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This project demonstrates the power of transfer learning and multi-task learning in affective computing, providing a solid foundation for emotion recognition applications.